Queue Delay Analysis in Network Congestion

Dijkstra vs A* vs Softmax Randomized Routing

Method summary

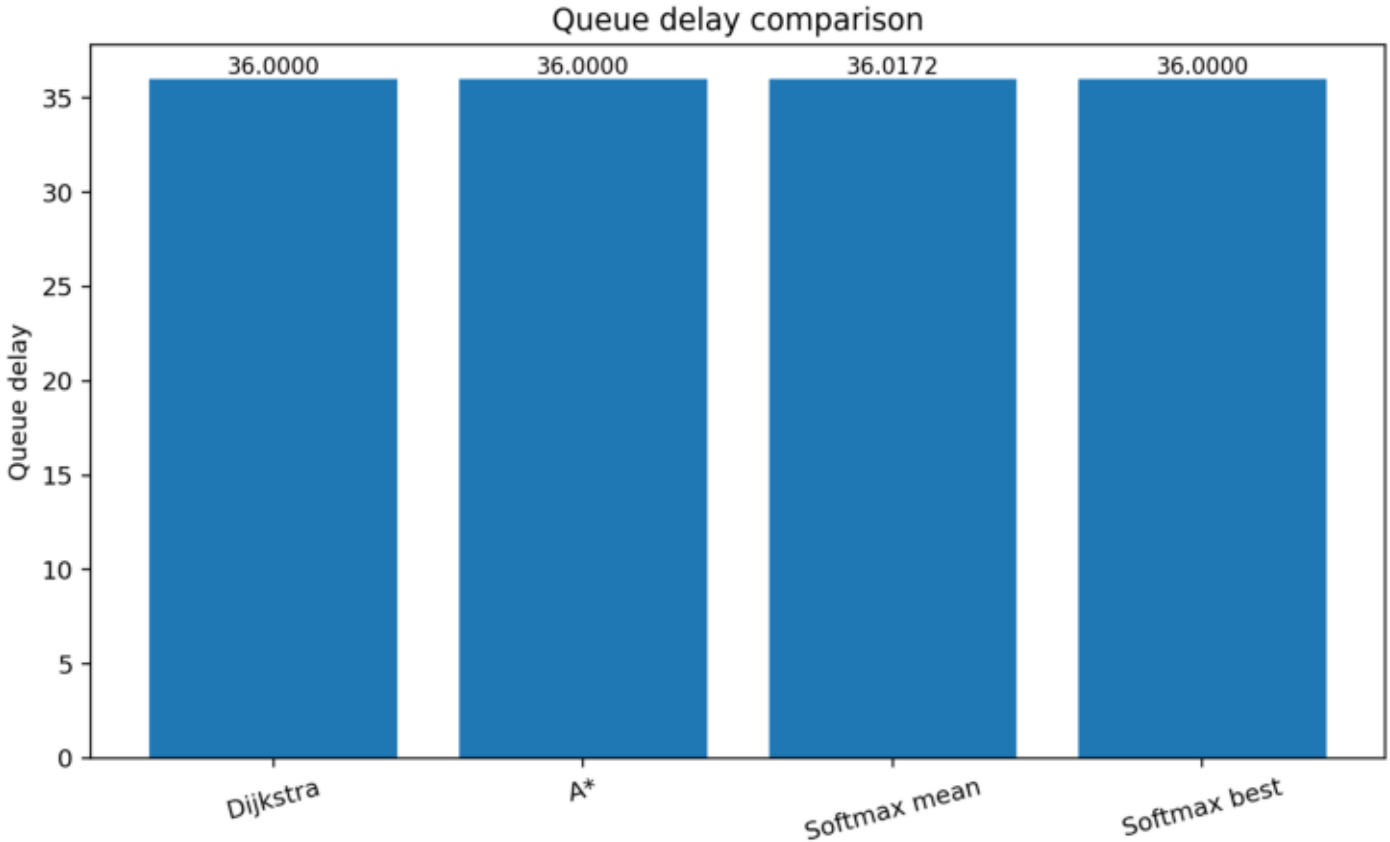
algorithm	best_delay	mean_delay	success_rate	expanded_nodes	runtime_ms	path_length
Dijkstra	36.0	36.00000	1.0	84.000	0.091095	19
A*	36.0	36.00000	1.0	76.000	0.098943	19
Softmax Randomized (T=0.15)	36.0	36.01725	1.0	18.000	115.626981	19
Softmax Randomized (T=0.3)	36.0	36.08750	1.0	18.005	123.340650	19
Softmax Randomized (T=0.5)	36.0	36.26825	1.0	18.059	118.767696	19
Softmax Randomized (T=0.8)	36.0	38.33700	1.0	18.728	122.422879	19
Softmax Randomized (T=1.2)	36.0	44.32075	1.0	20.554	144.109968	19

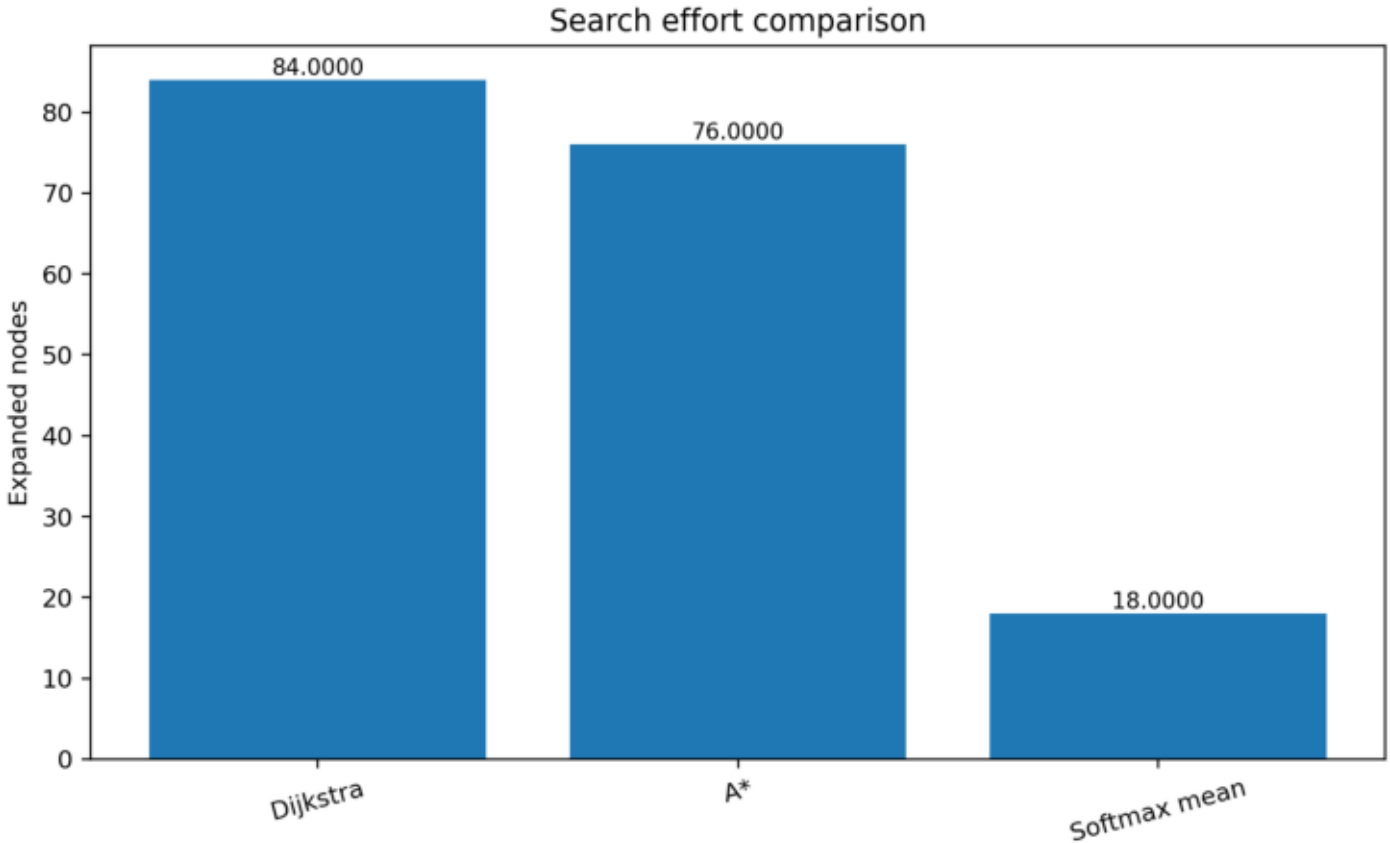
Temperature sensitivity

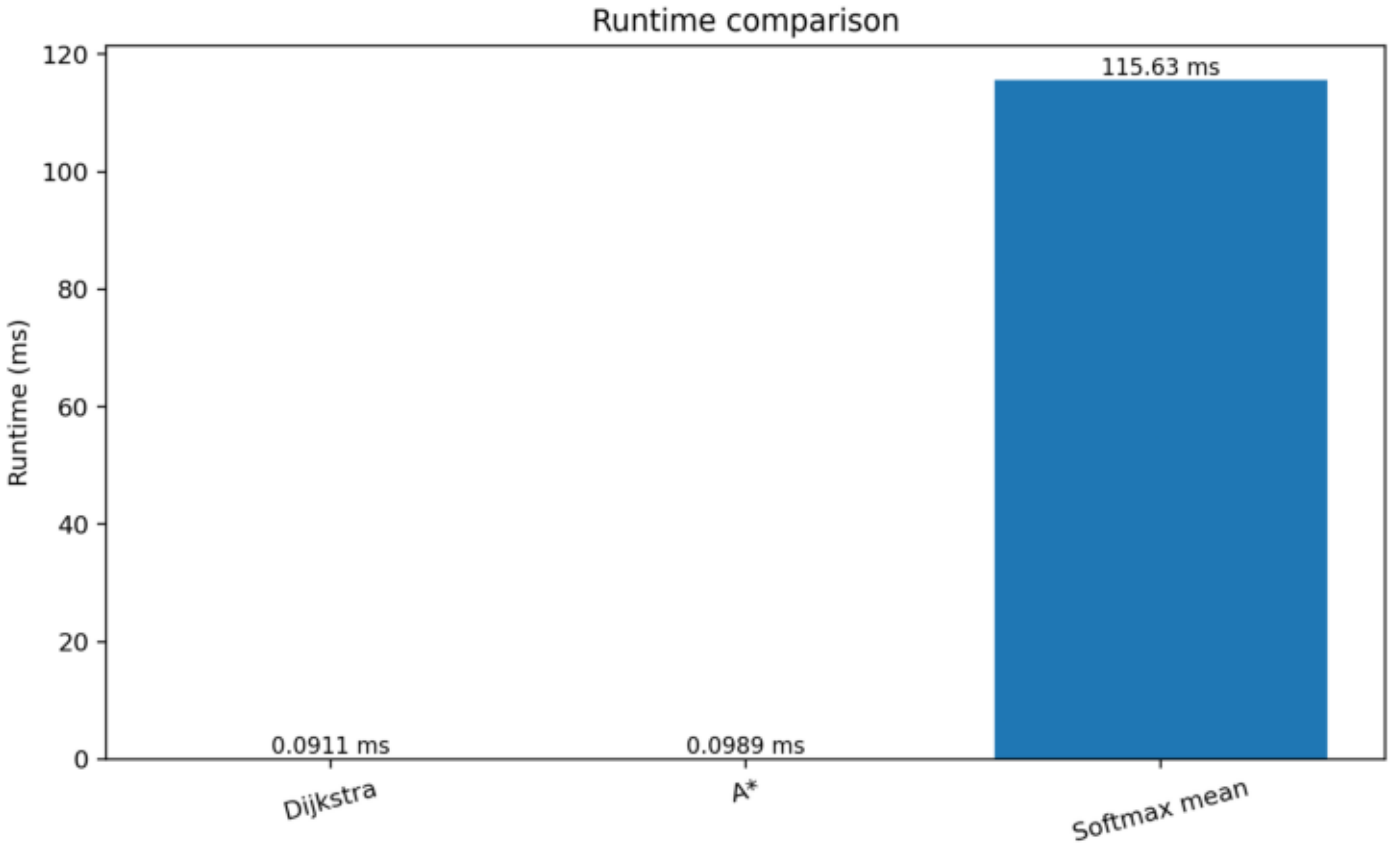
temperature	best_delay	mean_delay	std_delay	success_rate
0.15	36.0	36.01725	0.091255	1.0
0.30	36.0	36.08750	0.237368	1.0
0.50	36.0	36.26825	0.790359	1.0
0.80	36.0	38.33700	15.037833	1.0
1.20	36.0	44.32075	28.207242	1.0

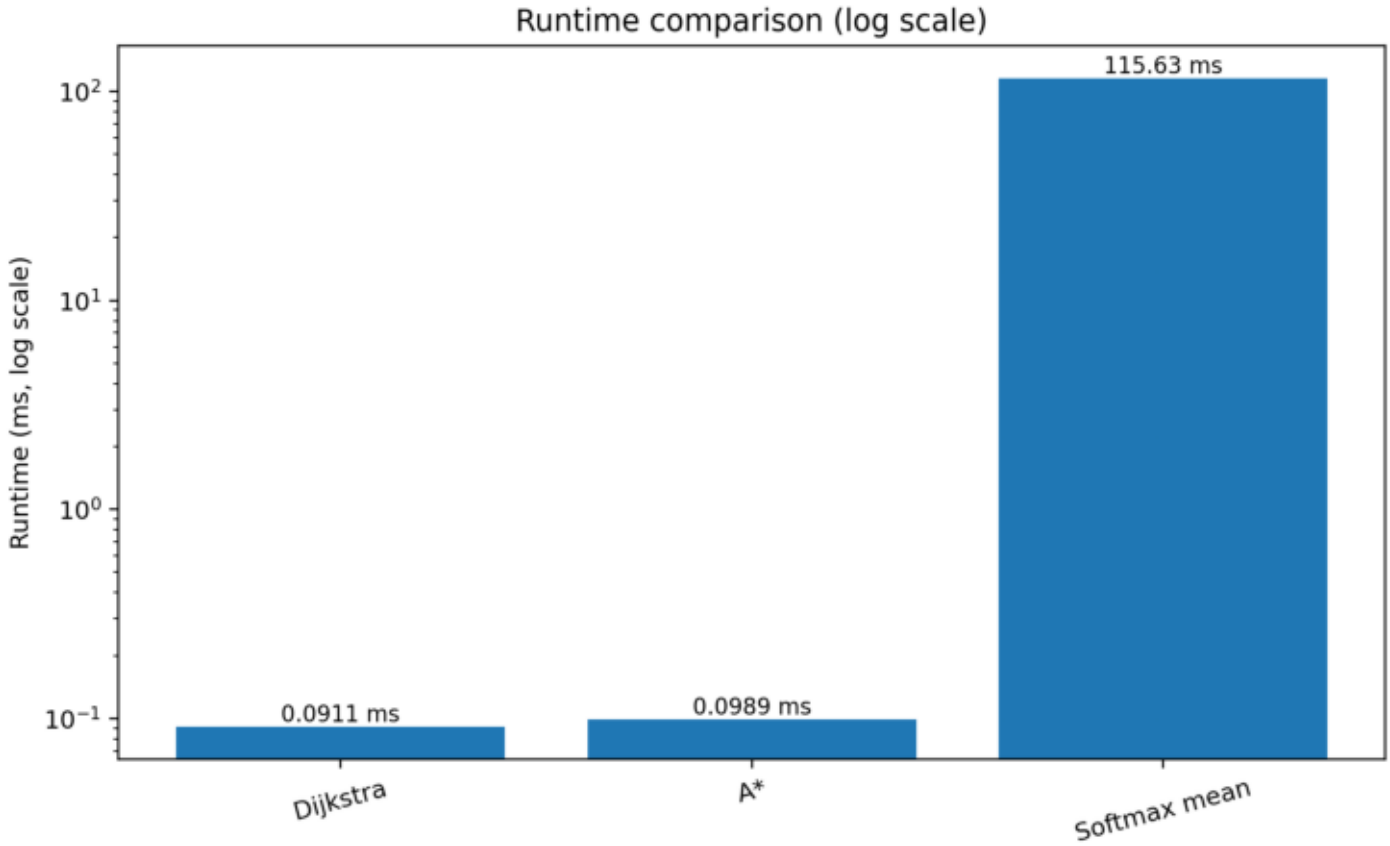
Inference

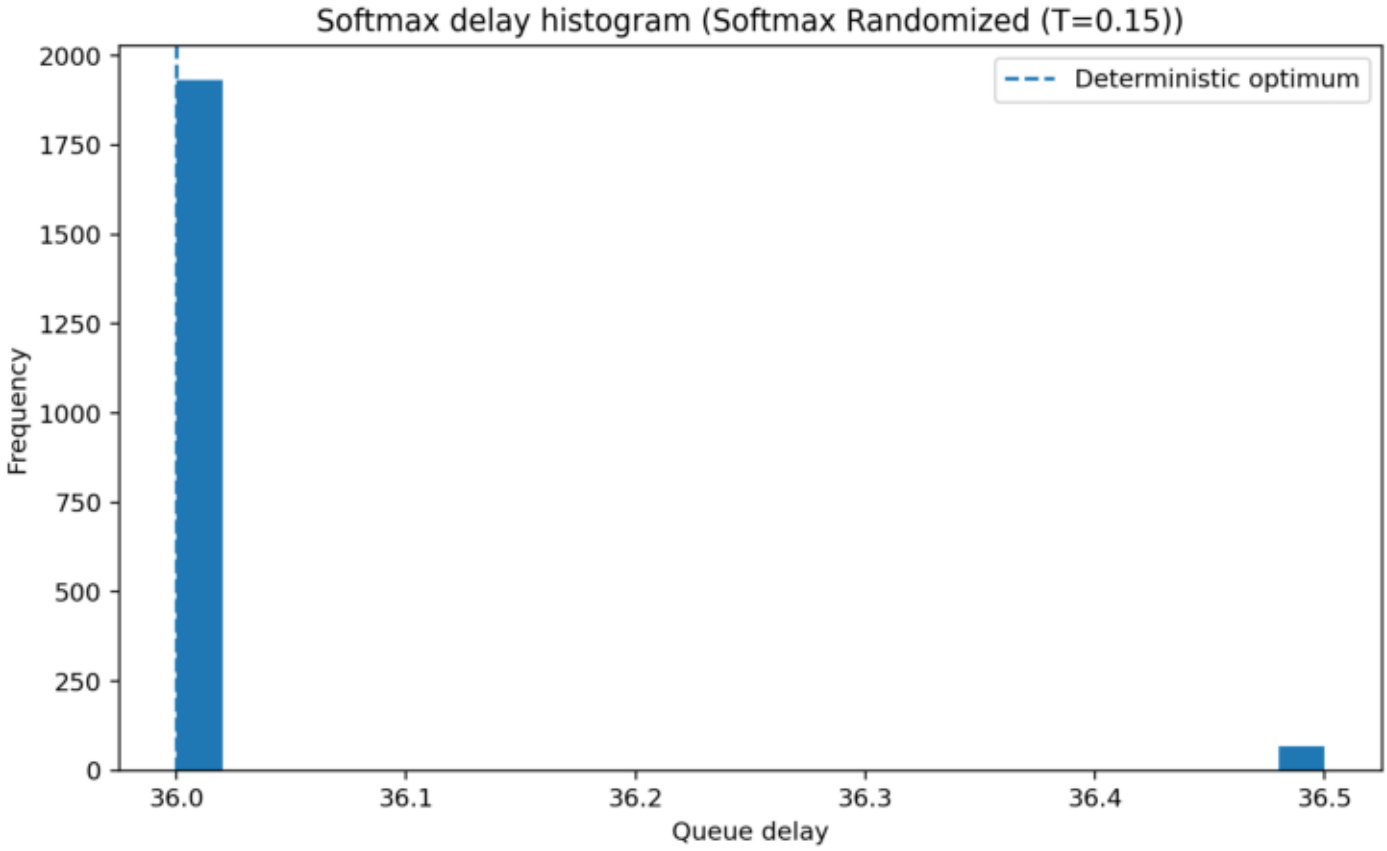
Dijkstra gives the exact minimum queue-delay path with total delay 36.0000. A* reaches the same optimum in this graph, which shows that the heuristic guidance is compatible with the network layout for this instance. The softmax randomized method introduces controlled stochastic exploration: instead of always choosing the minimum-energy next hop, it samples the next node using a softmax distribution over candidate energies. Among the tested temperatures, Softmax Randomized (T=0.15) gave the strongest average result with mean delay 36.0172, best observed delay 36.0000, and success rate 100.00%. Its average delay is 0.0479% above the deterministic optimum, which is expected because randomness trades perfect optimality for route diversity and exploration. The histogram acts as an empirical PDF approximation and shows the concentration of randomized delays, while the empirical CDF shows the probability of obtaining a route below a chosen threshold. For this graph, the probability of getting a path within 10% of the deterministic optimum is 100.00%. Therefore, Dijkstra is the best choice when exact minimum queue delay is required, A* is a strong informed deterministic alternative, and softmax randomized routing is useful when the study needs probabilistic behavior, multiple candidate routes, or an algorithmic base that can later be extended to dynamic congestion scenarios.

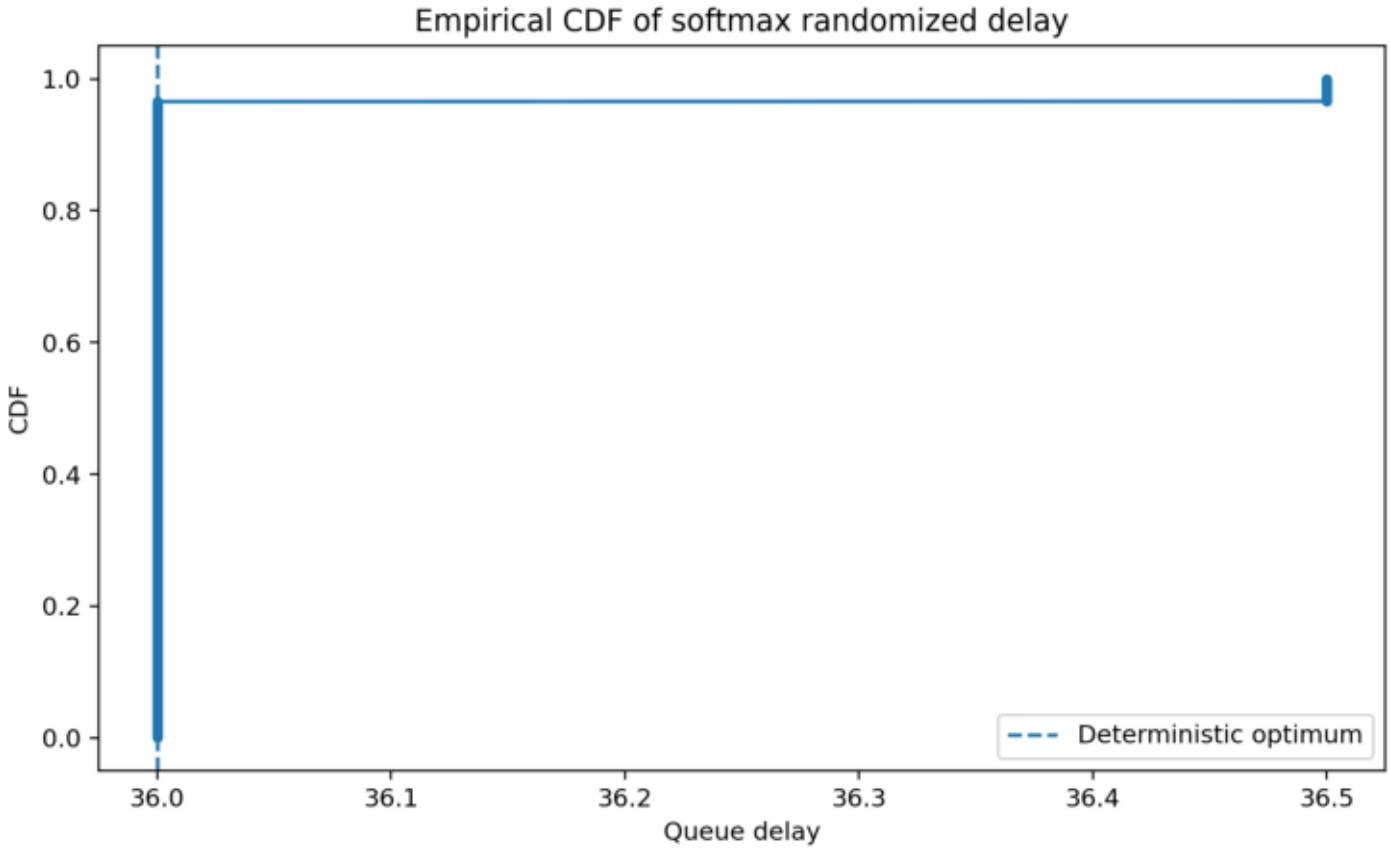

Queue delay comparison
36.0000
36.0000
36.0172
36.0000
Queue delay
0
5
10
15
20
25
30
35
Dijkstra
A*
Softmax mean
Softmax best

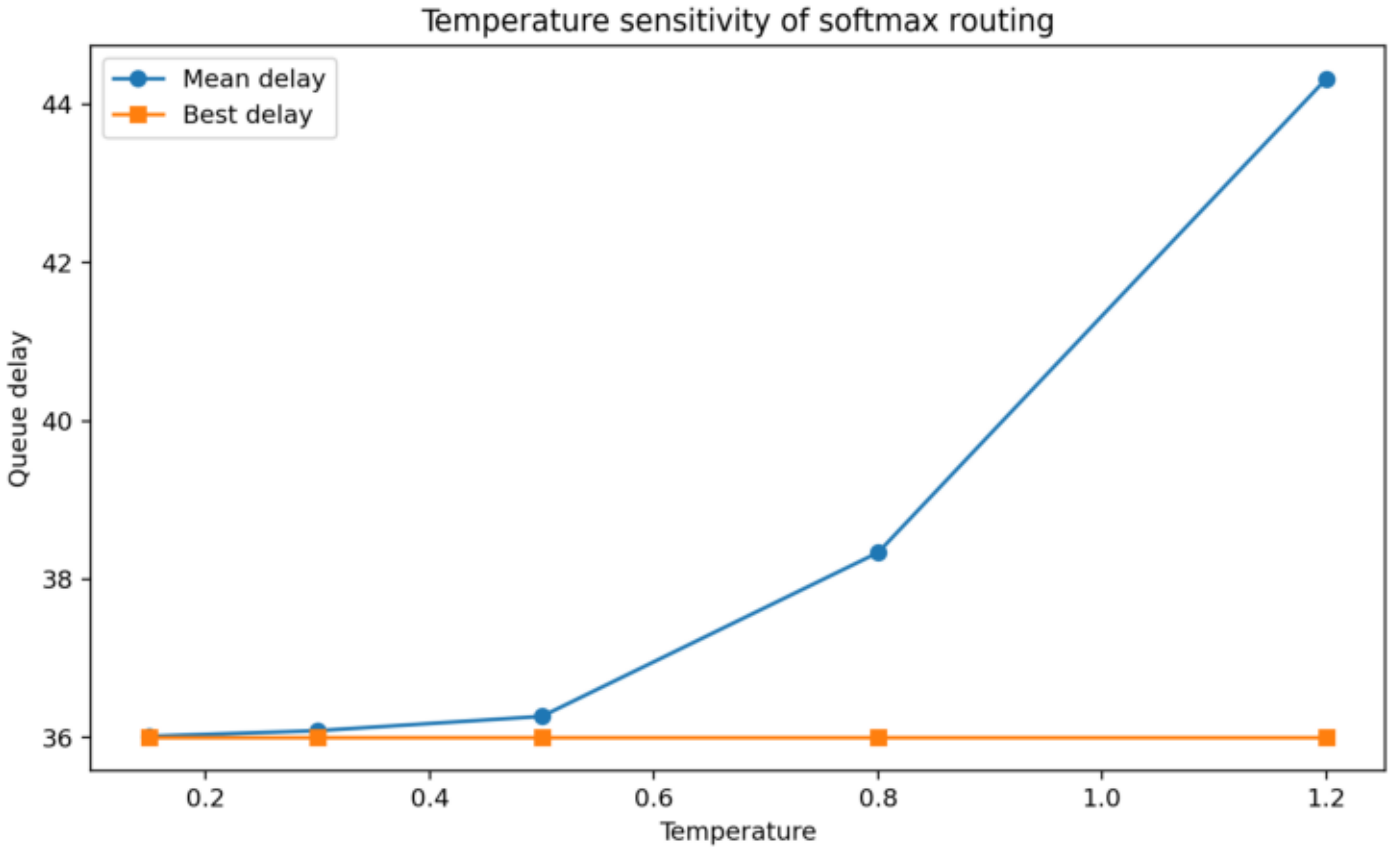

Search effort comparison
Expanded nodes
0
10
20
30
40
50
60
70
80
84.0000
76.0000
18.0000
Dijkstra
A*
Softmax mean


Runtime comparison
Runtime (ms)
120
100
80
60
40
20
0
115.63 ms
0.0911 ms
0.0989 ms
Dijkstra
A*
Softmax mean


Runtime comparison (log scale)
Runtime (ms, log scale)
10^2
10^1
10^0
10^{-1}
115.63 ms
0.0911 ms
0.0989 ms
Dijkstra
A*
Softmax mean


Softmax delay histogram (Softmax Randomized (T=0.15))
Deterministic optimum
Frequency
2000
1750
1500
1250
1000
750
500
250
0
36.0
36.1
36.2
36.3
36.4
36.5
Queue delay


Empirical CDF of softmax randomized delay
1.0
0.8
0.6
0.4
0.2
0.0
CDF
36.0
36.1
36.2
36.3
36.4
36.5
Queue delay
Deterministic optimum


Temperature sensitivity of softmax routing
Mean delay
Best delay
Queue delay
44
42
40
38
36
0.2
0.4
0.6
0.8
1.0
1.2
Temperature